

CNS

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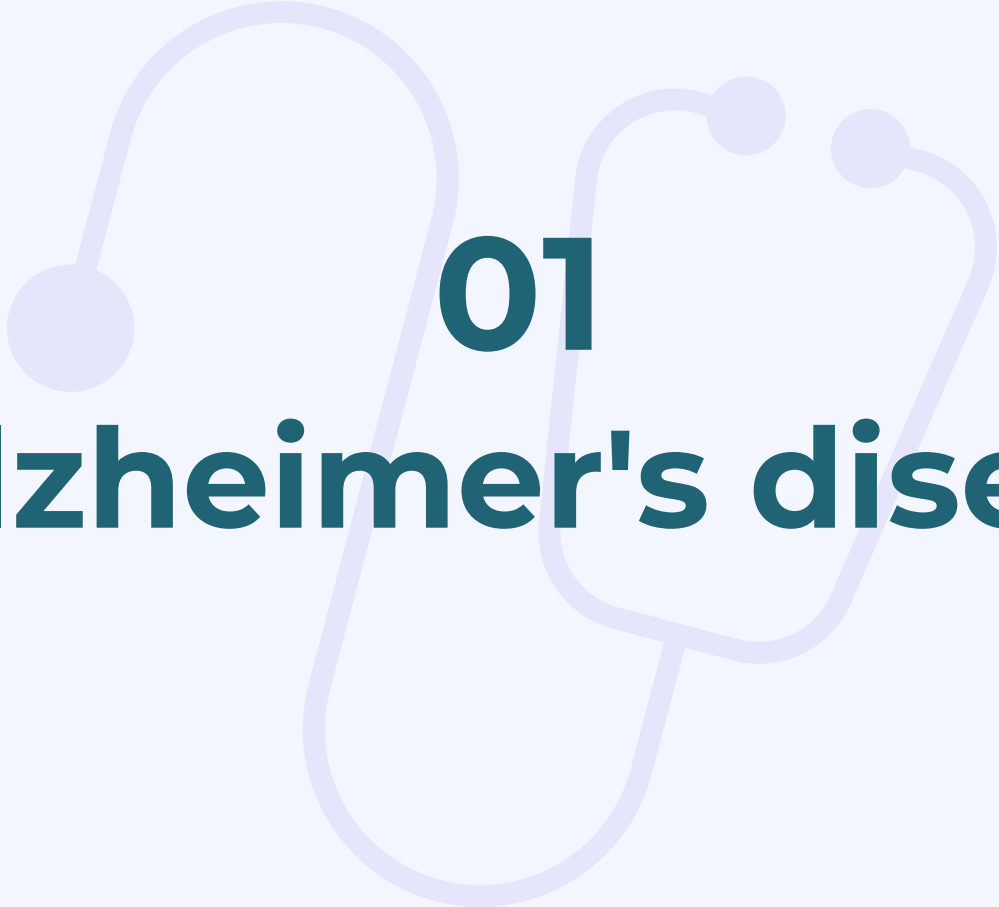
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A large, light purple outline of a human brain is centered in the background. A stethoscope is draped over the brain, with its chest piece resting on the right side and its ear pieces extending towards the top. The stethoscope is a darker shade of purple.

01

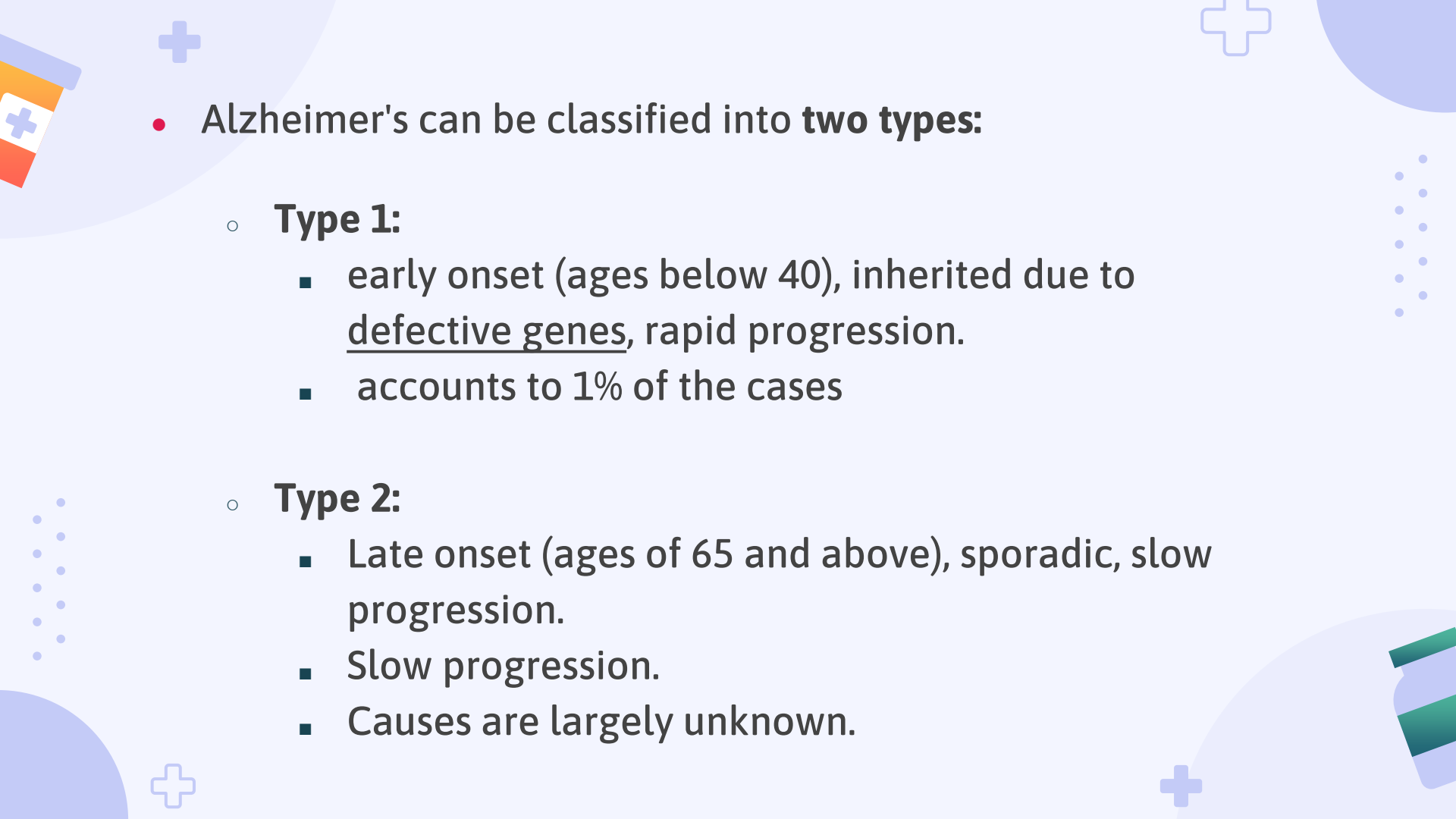
Alzheimer's disease



Introduction

- **Alzheimer's** disease is a neurodegenerative disease (irreversible damage as affected neurons can not regenerate) characterized by:
 - Progressive **M**emory loss.
 - **B**ehavioral abnormalities (Dementia) : depression ,aggression, motor hyperactivity.
 - Progressive **C**ognitive impairment: unable to recognize people and in later stages concepts.



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- Alzheimer's can be classified into **two types**:

- **Type 1:**

- early onset (ages below 40), inherited due to defective genes, rapid progression.
- accounts to 1% of the cases

- **Type 2:**

- Late onset (ages of 65 and above), sporadic, slow progression.
- Slow progression.
- Causes are largely unknown.

pathology

On the brain:

- **Gross brain atrophy, with significant reduction in hippocampus** (plays a role in memory function)

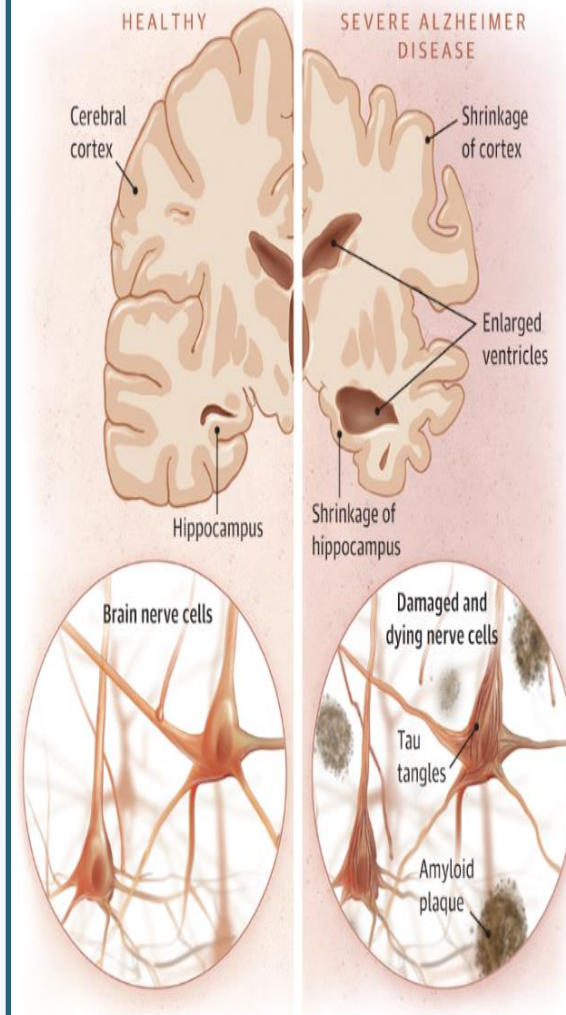
On neurons:

- Formation of **β - amyloid plaques** which reduces neuron connection.
- Formation of neurofibrillary tangles (tau tangles) which cause further degeneration of neurons.

Biochemistry:

- Decreased acetylcholine (an important neurotransmitter in memory function).
- Increase of glutamate (an excitatory neurotransmitter which causes neuron damage in Alzheimer's disease).

Brain changes in Alzheimer disease



Treatment

Alzheimer's is not curable, medication only serves to slow the progression of disease and enhance cognitive functions, however, total memory loss and worsening of patient cognitive functioning is inevitable, failing to recognize objects, concepts, and even loved ones.

- **Alzheimer's is a disease of the family** and impacts not only the patient but his family and caregiver as mental state deteriorates.

- **Treatment available are:**

- Cholinesterase inhibitors (increase Ach)
- NMDA blockers (block the action of glutamate)

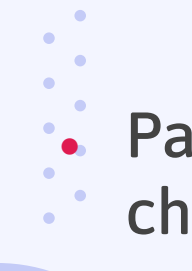




Cholinesterase Inhibitors (First line) eg: Donepezil, Rivastigmine, Galantamine.

- **M.O.A:**

- Selective Inhibition of Acetyl cholinesterase enzyme (AChE) in the brain and improvement of cholinergic transmission & Memory.



- Patients with Alzheimer's disease who treated with anti-cholinesterase drugs show improvement in memory and cognition symptoms by 10%.



- **Adverse effects include:**

- Selectivity on cerebral AChE is not absolute.
- Bradycardia.
- Salivation and Urination.
- Diarrhea and abdominal spasms.

- **Drug interactions:**

- All these drugs are substrates to CYP-450 except rivastigmine and so clinically significant drug interactions are present with drugs that may inhibit CYP-450.
- The use of drugs that block Ach receptor centrally (TCA anti-depressants) will negatively impact any improvements of AChE.I



Brands



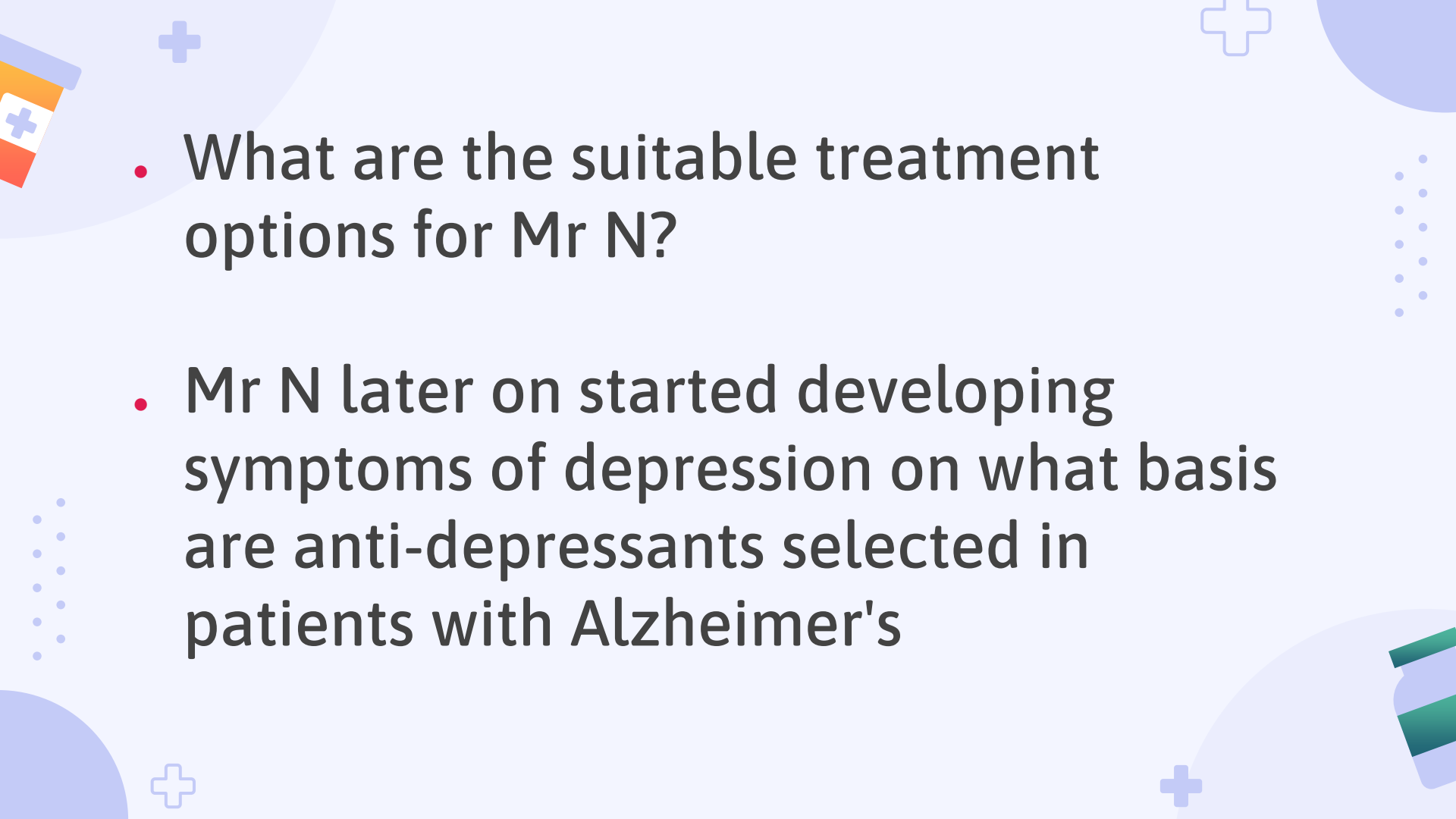
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- **NMDA(N-Methyl-D-Aspartate) Receptor antagonists (Second line) eg: Memantine**
 - NMDA-receptor antagonists are usually used as Add-on drug in case of Severe and rapid progression of Alzheimer's Disease.
 - **M.O.A**
 - Blocking of NMDA-receptors (glutamate receptors) thus reducing its excitatory effect.
 - **Adverse effects include:** Insomnia, agitation, confusion and headache.
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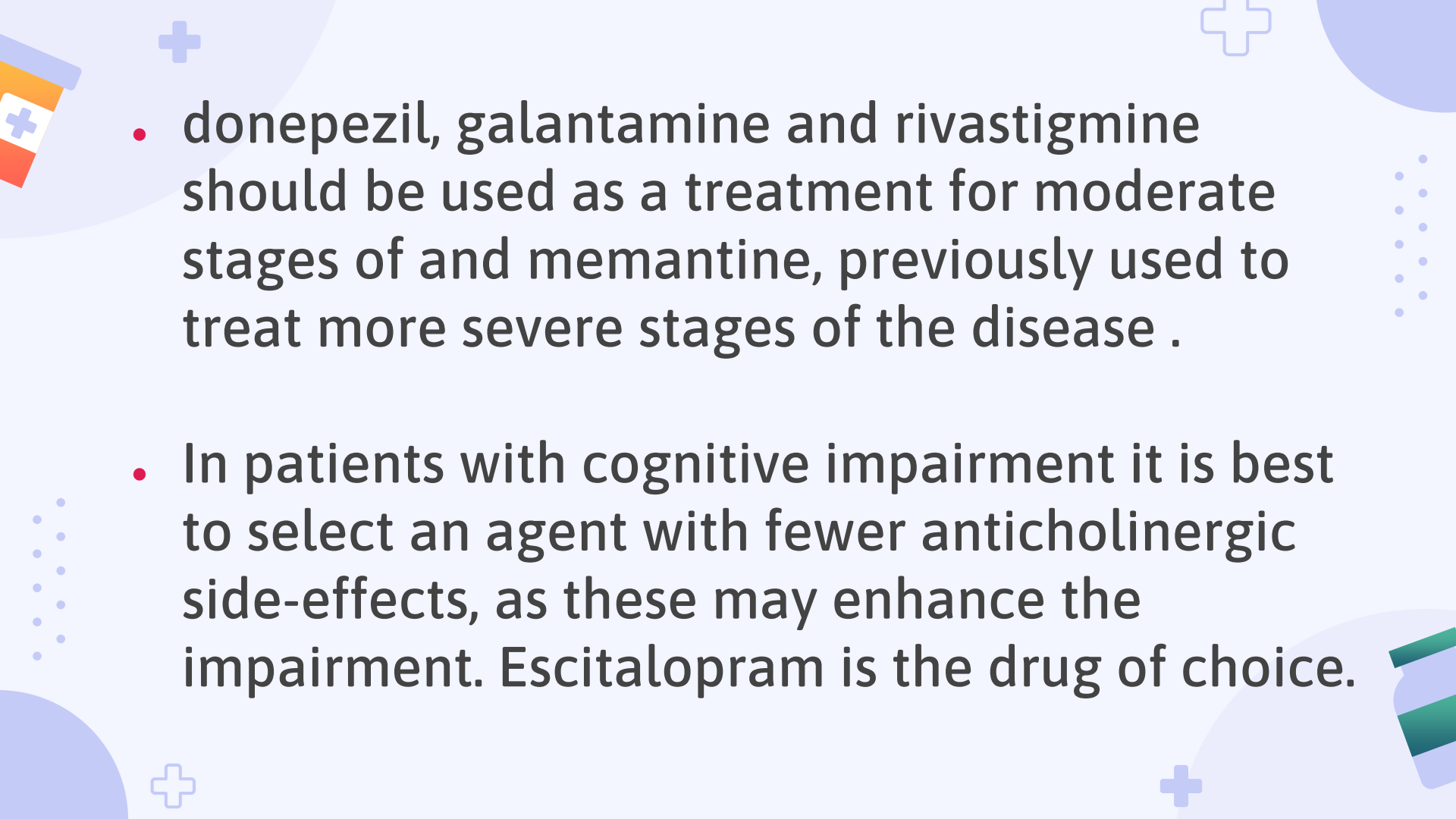
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Cases

- The wife of Mr N , a 68-year-old black Caribbean male, is asking for some advice from the pharmacist because she is concerned about the changes in behavior her husband has been showing over the last few months. In contrast with Mr N's previous personality, his mood is now very volatile and somewhat unpredictable, his memory and concentration seem to be very poor and, at times, he may be very impulsive indeed. For the first time in his life he is now using words and language expressions that are both rude and vulgar .

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- What are the suitable treatment options for Mr N?
 - Mr N later on started developing symptoms of depression on what basis are anti-depressants selected in patients with Alzheimer's

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- donepezil, galantamine and rivastigmine should be used as a treatment for moderate stages of and memantine, previously used to treat more severe stages of the disease .
 - In patients with cognitive impairment it is best to select an agent with fewer anticholinergic side-effects, as these may enhance the impairment. Escitalopram is the drug of choice.

Cases

- Modest improvement in the memory of patients with Alzheimer's disease may occur with drugs that increase transmission at which of the following receptors?
 - A. Adrenergic.
 - B. Cholinergic.
 - C. Dopaminergic.
 - D. GABAergic.
 - E. Serotonergic.

Answers

- Correct answer = B. AChE inhibitors, such as rivastigmine, increase cholinergic transmission in the CNS and may cause a modest delay in the progression of Alzheimer's disease. Increased transmission at the other types of receptors listed does not result in improved memory.

Cases

- Which medication is a glutamate receptor antagonist that can be used in combination with an acetylcholinesterase inhibitor to manage the symptoms of Alzheimer's disease?
 - A. Rivastigmine.
 - B. Ropinirole.
 - C. Fluoxetine.
 - D. Memantine.
 - E. Donepezil.

Answers

- Correct answer = D. When combined with an acetylcholinesterase inhibitor, memantine has modest efficacy in keeping patients with Alzheimer's disease at or above baseline for at least 6 months and may delay disease progression.